

Hall Ticket No.

Code: R122132

Stanley College of Engineering & Technology for Women (A)

B.E III Semester (Main) Examinations - January-2023

***ECE & EEE*-Probability Theory & Stochastic Processes**

Time: 03 Hours

Max. Marks-60

- Note:** i). First Question is Compulsory. Answer any Four out of remaining Six questions.
ii) Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii) Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. Define the Probability? What are the axioms of probability? [2M CO1 BTL1]
- b. Find out the probability of getting a 2 or a 5 when a die is rolled? [2M CO3 BTL1]
- c. State the Central Limit Theorem with an example [2M CO2 BTL2]
- d. Summarize the different measures of Central Tendency [2M CO3 BTL2]
- e. Distinguish between ensemble averages and the time averages of random process? [2M CO4 BTL2]
- f. Write the equations of Wiener-Khinchine relations? [2M CO5 BTL2]

Part-B

4X12=48M

2. a) State and prove the Baye's theorem and Total probability Theorem? [8M CO1 BTL2]
- b) When two dice are thrown, determine the probabilities from axiom 3 for the following events. (i) $A = \{\text{Sum} = 7\}$ (ii) $B = \{8 \leq \text{Sum} \leq 11\}$ (iii) $C = \{10 < \text{Sum}\}$ (iv) $P(B \cap C)$ [4M CO1 BTL3]
3. a) Define the CDF? State and prove its properties [6M CO1 BTL4]
- b) A continuous random variable X defined by probability density function given by $f(x) = 5(1-x^4)/4$ $0 \leq x \leq 1$. Calculate $E[X]$, $E[X^2]$ and variance. [5M CO2 BTL3]
4. a) Explain how Moment generating function generates the moments? State and prove its properties? [7M CO2 BTL4]
- b) The joint probability density function of $f(x,y)$ is given by $f(x,y) = 8xy$ $0 \leq x \leq 1$, $0 \leq y \leq x$
 - (i) Determine the marginal density of X and Y.
 - (ii) Determine the conditional density functions of X and Y. Verify that whether X and Y are independent. [5M CO2 BTL4]

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5. a) Fit a parabola to the following data

X	-2	-1	0	1	2
Y	29	25	22	20	19

[7M CO3 BTL4]

b) Fit a second degree polynomial for the following data by the method of least Squares

x	10	12	15	23	20
y	14	17	23	25	21

[5M CO3 BTL6]

6. a) Categorize random processes into first order ,second order , wide- sense and strict-sense stationary based on their characteristics and properties.

[6M CO4 BTL4]

b) Assess whether the random process $X(t)=A\cos(\omega_0 t+\Theta)$ is wide stationary or not, where A, ω_0 are constants and Θ is a uniformly distributed random variable on the interval $(0,2\Pi)$

[6M CO4 BTL5]

7. a) Define and prove any Four properties of PSD of random process. [6M CO5 BTL4]

b) Determine the auto correlation function for the following power density spectrum:

$$S_{xx}(\omega) = \frac{157+12\omega^2}{(16+\omega^2)(9+\omega^2)}$$

[6M CO5 BTL3]

Hall Ticket No. -

Code: R1223F

Stanley College of Engineering & Technology for Women (A)

B.E (ECE) III Semester (Main) Examinations-January-2023

Managerial Economics and Accountancy

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. What is time perspective? [2M CO1 BT1]
- b. What is income elasticity of demand? [2M CO2 BT1]
- c. Define isoquants. [2M CO3 BT1]
- d. What is double entry system? [2M CO4 BT1]
- e. Define journal [2M CO4 BT1]
- f. Define payback period with formulae. [2M CO5 BT2]

Part-B

4X12=48M

2. a) Discuss how Managerial Economics is useful to Engineers? [8M CO1 BT5]
- b) Discuss briefly (a) Opportunity cost (b) Discounting Principle [4M CO1 BT5]
3. a) What are the exceptions of Law of demand? [8M CO2 BT3]
- b) What is demand forecasting and explain its qualitative techniques. [4M CO2 BT3]
4. a) Explain Break-even analysis and what are its uses. [8M CO3 BT6]
- b) List out the features of Perfect competition. [4M CO3 BT6]
5. a) Mention advantages and Disadvantages of Ratio analysis. [8M CO4 BT5]
- b) Write importance of Petty cash book? [4M CO4 BT5]

6. A firm with cost of capital of 10% is considering two mutually exclusive projects A and

B, the details of which are:

[12M CO5 BT4]

	Project – A Rs.	Project – B Rs.
Investment	90,000	90,000
Cash flows Year– 1	12,000	52,000
2	22,000	42,000
3	32,000	22,000
4	42,000	12,000
5	52,000	12,000

Compute Pay Back Period and Net Present Value and suggest which project is acceptable

7. a) List out factors determining the working capital requirements.

[8M CO5 BTL2]

b) Write about (i) Gross working capital (ii) Networking capital

[4M CO5 BTL2]

Stanley College of Engineering & Technology for Women (A)**B.E (ECE) III Semester (Main) Examinations-January-2023****Digital System Design****Time: 03 Hours****Max. Marks-60**

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
 - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 - Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

- What do you understand by the number systems. Enlist the types of number systems with an example for each. [2M CO1 BTL1]
 - State and Prove Demorgan's Theorem [2M CO1 BTL1]
 - Describe Minterm. Write all Minterms for 3 Variables. [2M CO2 BTL2]
 - Describe Combinational and Sequential logic circuits. [2M CO3 BTL2]
 - What do you mean by race around condition, where it occurs and how to avoid it? [2M CO4 BTL1]
 - Define different modeling styles in Verilog HDL. [2M CO5 BTL1]

Part-B**4X12=48M**

- Convert the following
 - $(327.89)_{10} = (?)_{BCD}$
 - $(273.11)_{10} = (?)_2$
 - $(1111101.1101)_2 = (?)_8$
 - Describe the following with example.
 - Excess-3 code
 - Gray code
 - Weighted binary code
 - Perform the following subtractions using 2's complement method
 - $01100-00011$
 - $01000-010011$
- Given the logic function $f = A \bar{B} D + A B C + A \bar{B} C$
 - Make a truth table
 - Simplify using K map
 - Realize using only NAND gates.
 - Design a BCD to Excess-3 code converter. Realize using basic gates and also using only NAND gates.

4. a) Design and implement a full adder using two half adders and an OR gate.

[6M CO3 BTL2]

- b) Implement the full subtractor using IC 74138.

[6M CO3 BTL6]

5. a) Explain the working of master slave JK flipflop. Covert JK flipflop into T flipflops.

[6M CO4 BTL5]

- b) Design Mod-6 Asynchronous counter using JK flipflop.

[6M CO4 BTL6]

6. a) Explain VLSI design Flow with suitable flow diagram.

[4M CO5 BTL5]

- b) Write and Explain Verilog code for implementing 8 to 3 encoder.

[8M CO5 BTL5]

7. Minimize the following function using Quine-Mc Cluskey Tabular Method

$$F(A, B, C, D) = \sum M(0, 1, 2, 5, 6, 7, 8, 9, 10, 14)$$

[12M CO2 BTL5]

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Code: R122137

Stanley College of Engineering & Technology for Women (A)

B.E (ECE) III Semester (Main) Examinations - January-2023

Electronic Devices and Circuits

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
 - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 - Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

- a. What is fermi level in N-type semiconductor? [2M CO1 BTL1]
b. What is Zener breakdown voltage [2M CO1 BTL1]
c. What is the need of rectifier? [2M CO2 BTL1]
d. Define operating point. [2M CO3 BTL1]
e. Compare JFET with BJT. [2M CO4 BTL2]
f. Sketch a common source MOSFET amplifier [2M CO5 BTL1]

Part-B

4X12=48M

- a) Draw and explain V-I characteristics of PN diode. [6M CO1 BTL2]
b) Explain the construction and working of Zener diode
- a) Draw the diagram of full-wave rectifier with four diodes and explain its operation? [6M CO2 BTL2]
b) Derive the equation for ripple factor of half wave rectifier with C-filter. [6M CO2 BTL4]
- a) Give the list of different filters used in rectifier and their merits and demerits. [6M CO2 BTL3]
b) What is the Hall Effect? Derive an Expression for Hall Coefficient? [6M CO1 BTL5]
- a) Explain the working of NPN transistor. [6M CO3 BTL2]
b) Write the differences between CB, CE, and CC Amplifier Configurations. [6M CO3 BTL3]
- a) Describe the analysis of Transistor Amplifier circuit using h-parameters. [6M CO4 BTL4]
b) Explain the analysis of BJT amplifiers using exact and approximate model for CE [6M CO4 BTL5]
- a) Write short notes on Small Signal Model of JFET. [6M CO5 BTL6]
b) Explain the construction and working of Enhancement MOSFET. [6M CO5 BTL6]

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Stanley College of Engineering & Technology for Women (A)**B.E (ECE) III Semester (Main) Examinations - January-2023****Electromagnetic Theory and Transmission Lines****Time: 03 Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. State Gauss law for electric fields. [2M CO1 BTL2]
 b. Explain Biot-Savart's law. [2M CO1 BTL5]
 c. Define skin depth. [2M CO3 BTL1]
 d. Explain about inductance. [2M CO2 BTL5]
 e. Define phase velocity and group velocity. [2M CO4 BTL1]
 f. What are the applications of Smith Chart? [2M CO5 BTL2]

Part-B**4X12=48M**

2. a) State and prove Coulomb's law. [7M CO1 BTL1]
 b) Point charges 1mc and -2mc are located at $(3, 2, 1)$ and $(-1, 4)$ respectively. Calculate the electric force on a 10nc charge at $(0, 3, 1)$ and the electric field intensity at that point. [5M CO1 BTL3]
3. a) Write Maxwell's equations in point form and integral form. [8M CO2 BTL4]
 b) State Faraday's law of electromagnetic induction. [4M CO2 BTL1]
4. a) Derive the wave equation for conducting medium. [8M CO3 BTL2]
 b) Discuss about polarization. [4M CO3 BTL6]
5. a) State and prove Poynting theorem. [8M CO3 BTL1]
 b) Define Brewster angle and critical angle. [4M CO3 BTL1]
6. a) Define primary and secondary constants. [7M CO4 BTL1]
 b) Distinguish between open and short circuited transmission lines. [5M CO4 BTL4]
7. a) Define the reflection coefficient and Standing wave ratio. [6M CO5 BTL1]
 b) Calculate reflection coefficient and VSWR for a 50Ω line terminated with Matched load, Short circuit and Open circuit. [6M CO5 BTL3]

Stanley College of Engineering and Technology for Women(A)**B.E(EEE,ECE & IT) II Semester (Backlog) Examinations January-2023****Indian Constitution**

Time: 3 hours

Max.Marks:60

PART-A**5 X 2=10M****Note: Answer all questions(Compulsory)**

- | | |
|---|------------|
| 1 Who are the members of Drafting Committee? | 2 M CO1 L1 |
| 2 What are Panchayati Raj Institutions?Name them. | 2 M CO2 L2 |
| 3 Explain about Article-19. | 2 M CO3 L2 |
| 4 Why is NITI Aayog Necessary? | 2 M CO4 L1 |
| 5 What is the Composition of National Commission for Women? | 2 M CO5 L3 |

PART-B**5 X 10= 50M****Note: Answer all questions**

- | | |
|---|-------------|
| 6 a. Discuss the Significance of Indian Constitution. | 5 M CO1 L4 |
| b. What are the Features of 1935 Act? | 5 M CO1 L1 |
| OR | |
| c. Explain the Philosophical Foundations to the Indian Constitution. | 10 M CO1 L3 |
| 7 a. Examine the Powers and Functions of Prime Minister. | 5 M CO2 L4 |
| b. Explain about Emergency powers of President of India. | 5 M CO2 L2 |
| OR | |
| c. Write about Evolution of Local Government in India. | 10 M CO2 L4 |
| 8 a. Explain the Importance of Fundamental duties. | 5 M CO3 L3 |
| b. Briefly explain about DPSP. | 5 M CO3 L4 |
| OR | |
| c. Explain the features of Fundamental rights. | 5 M CO3 L2 |
| d. Explain the Five Writs in Indian Constitution. | 5 M CO1 L1 |
| 9 a. Explain about Composition of Inter State Council. | 5 M CO4 L3 |
| b. Write an essay on Role and Functions of Finance commission of India. | 5 M CO4 L4 |
| OR | |
| c. Explain about the functions of NITI Aayog. | 5 M CO4 L2 |
| d. Explain about Administrative relations of Union-State. | 5 M CO4 L4 |
| 10 a. What are the Powers and Functions of NHRC? | 10 M CO5 L4 |
| OR | |
| c. What are the Powers and Functions of National Commission for Women? | 10 M CO5 L1 |

Stanley College of Engineering and Technology for Women(A)**B.E (ECE, EEE & IT) II Semester (Backlog) Examinations January-2023****Essence of Indian Traditional Knowledge****Time: 3 hours****Max.Marks:60****PART-A****Note: Answer all questions (Compulsory)****5 X 2=10M**

- 1 Define Civilization. 2M L1
- 2 What are Brahmanas and Aryankas? 2M L1
- 3 List the Heterodox schools of Indian Philosophy. 2M L1
- 4 Write a short note on Sama Veda. 2M L1
- 5 What were the subjects of study in Ancient India? 2M L1

PART-B**Note: Answer all questions****5 X 10= 50M**

- 6 a. Discuss the general characteristics of Culture. 5M L2
- b. Explain the uniqueness of Indian Culture. 5M L2

OR

- c. Write about ancient India. 5M L1
- d. Write an essay on the influence of the English in Modern India. 5M L1
- 7 a. Summarize the developments of South Indian Languages and Literature. 5M L2
- b. Explain about the Vedas and Upanishads. 5M L2

OR

- c. Discuss the classic Sanskrit literature. 5M L2
- d. Discuss the literature of North Indian Languages. 5M L2
- 8 a. Examine the salient features of orthodox schools of Indian Philosophy. 5M L2
- b. What is the important role played by the Bhakti and Sufi religious reform movements in India? 5M L2

OR

- c. What is Sunyavada of Buddhism? Explain. 5M L2
- d. Discuss the Religious reform movements in Modern India. 5M L2
- 9 a. Write an essay on "Ancient Indian Architecture". 5M L2
- b. Explain elaborately about the Fine Arts of India. 5M L2

OR

- c. Discuss about Indian classical dance forms. 5M L2
- d. Justify the statement "Science and technology can solve all the world's problems, and historically it has been shown to make the world better and better." 5M L3
- 10 a. Write about the education system in Medieval India. 5M L1
- b. List any five contributions made by Ancient Indian scientists to the world of Science. 5M L1

OR

- c. Explain the Gurukula Education system. 5M L2
- d. Discuss the role of Education in social change. 5M L1

Hall Ticket No-

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Stanley College of Engineering and Technology for Women(A)

B.E (ECE & EEE) II Semester (Backlog) Examinations January-2023

Engineering Physics

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 A plane transmission grating having 6000 lines per cm is used in normal incidence mode. Calculate the angular separation between the two spectral lines of wavelengths 5890 Å and 5896 Å in second order. 2 M
- 2 Explain the phenomenon of total internal reflection. 2 M
- 3 Define a) Space lattice and b) Unit cell. 2 M
- 4 Calculate the wavelength of an electron accelerated to a potential of 900 V. 2 M
- 5 What is Lenz's law? 2 M

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. Explain the Young's double slit experiment to produce the interference pattern and obtain the conditions for maxima and minima. 5 M
- b. Discuss intensity expression for N slit diffraction grating. 5 M
- OR**
- c. Analyze interference of light by division of amplitude and division of wave front with examples. 5 M
- d. Classify Fresnel's and Fraunhofer's diffraction patterns. 5 M
- 7 a. Explain the construction and working of semiconductor laser. 5 M
- b. Explain the types of optical fibers basing on refractive index profiles. 5 M
- OR**
- c. Analyze the basic components required for producing laser 5 M
- d. Derive an expression for Numerical aperture and acceptance angle of an optical fiber. 5 M
- 8 a. Deduce the equation for the interplanar distance for cubic crystal. 5 M
- b. Classify the point defects in a crystal. 5 M
- OR**
- c. Formulate the expression for the concentration of Schottky defects in ionic crystals. 5 M
- d. What are Miller indices? How to obtain them for a set of lattice planes? 5 M
- 9 a. Formulate the Schrodinger's time independent wave equation. 5 M
- b. Explain polarization in dielectric materials. 5 M
- OR**
- c. Summarize the properties of wave function Ψ and explain its significance. 5 M
- d. Determine an expression for electric field due to continuous charge distribution. 5 M
- 10 a. Explain the concept of displacement current in dielectrics. 5 M
- b. Explain Faraday's law and Lenz's law. 5 M
- OR**
- c. Obtain Poynting theorem. 5 M
- d. Build the Maxwell's equations for static fields. 5 M

Hall Ticket No:

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Stanley College of Engineering and Technology for Women (A)

B.E (ECE) II Semester (Backlog) Examinations January-2023

Circuit Theory

Time: 03 Hrs

Max.Marks:60

PART A

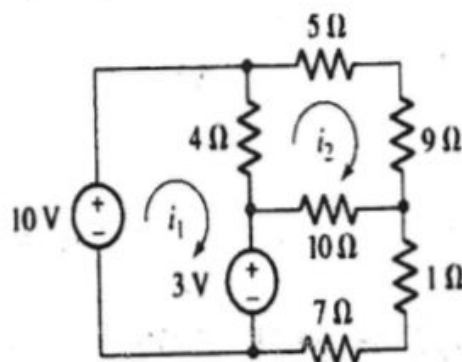
Note: Answer All Questions (Compulsory)

1. Define Kirchhoff's voltage and current law. 2M L1
2. Recognize the difference between steady state and transient response. 2M L1
3. Recall condition for symmetry and reciprocity in a two port network. 2M L1
4. At resonance, the current is maximum in a series circuit and minimum in a parallel circuit. Why? 2M L1
5. Describe the difference between active filters and passive filters. 2M L1

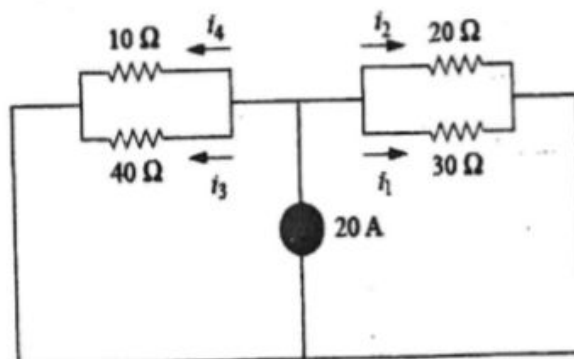
PART-B

Note: Answer all questions

- 6 a. Determine the currents i_1 and i_2 in the circuit shown using mesh analysis. 5M L3

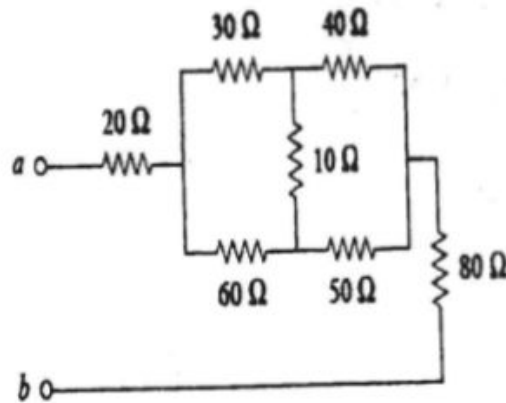


- b. By using current division rule calculate i_1 , i_2 , i_3 and i_4 in following circuit. 5M L3



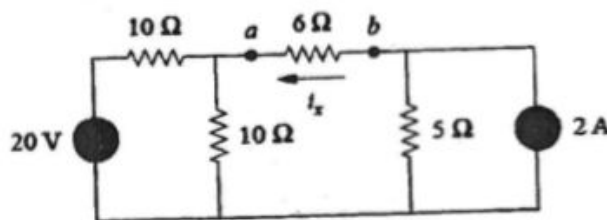
OR

c.

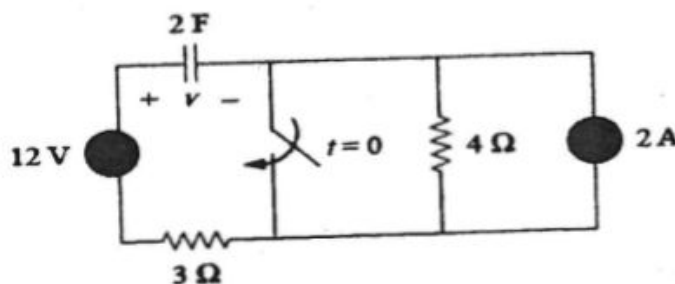
Find the resistance R_{ab} in circuit shown?

d. Find the Thevenin equivalent looking into terminals a-b in the circuit shown.

5M L3

7 a. Calculate the capacitor voltage for $t < 0$ and $t > 0$ for the circuit shown.

5M L3



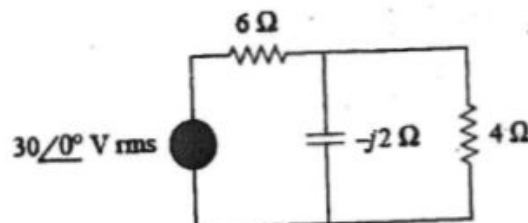
b. Outline a procedure with an example to formulate integro-differential equation for a series RLC circuit.

5M L4

OR

c. Calculate the average power delivered by source in following figure

5M L3



d. Illustrate with an example the complete response in a RC network.

5M L4

8 a. Create a two port network that satisfies following two equations.

5M L6

$$V_1 = 6V_2 - 4I_2$$

$$I_1 = 7V_2 - 2I_2$$

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- b. Measure transmission parameters of a two port network whose z parameters are $Z_{11}=6$ $Z_{12}=4$ $z_{21}=4$ $Z_{22}=6$ 5M L5

OR

- c. Develop a two port network that realizes the following z parameters. 5M L6

$$Z_{11}=20$$

$$Z_{12}=20$$

$$Z_{21}=5$$

$$Z_{22}=10$$

- d. Two two-port networks are connected in cascade. Recommend type of parameters such that the overall parameter matrix is the product of individual transmission parameter matrices and prove it. 5M L5

- 9.a. Explain concept of resonant frequency series RLC circuit 5M L2

- b. Analyze the concept of pole zero cancellation in impedance functions 5M L4

OR

- c. Explain admittance function with an example 5M L2

- d. Outline the differences between series and parallel resonant circuits 5M L4

10. a. Explain the Design of a m-derived low pass filter 5M L2

- b. Explain pass band, stop band and attenuation 5M L2

OR

- c. Explain the design of K-type band pass filters 5M L2

- d. Differentiate between constant K filters and m derived filters 5M L2

Hall Ticket No :

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Stanley College of Engineering and Technology for Women (A)

B.E(ECE,EEE & IT) II Semester (Backlog) Examinations January-2023

English

Time : 3 hours

Max.Marks : 60

PART-A

Note : Answer all questions (Compulsory)

5 X 2=10M

2 M BL1 CO1

1 Select the appropriate answer and fill in the blanks. (1/2m x 4 = 2 M)

- She does all her work _____ great care. (with/ by)
- I saw myself _____ the mirror. (in/on)
- The necklace was _____ present from her parents. (a/an/the)
- They were going to _____ concert. (Article) (a/ an/ the)

2 Use the appropriate tense forms of the given verbs and fill in the blanks.(1/2 m x4 = 2M) 2 M BL1 CO2

- How have you been _____ ? (do)
- He was badly wounded and _____ (bleed) profusely.
- I usually _____ (teach) Chemistry, but I _____ (teach) Biology this year.
- She _____ (finish) her breakfast and went to school.

3 Select the correct answer. (1/2m x 2 = 1 M)

2 M BL4 CO3

- A study of coins. _____ (numismatics/balistics)
 - Handwriting which is difficult to read. _____ (legible/illegible)
- Rewrite as directed.(1/2m x 2 = 1 M)
- Lavanya said, " We painted the house yesterday (Indirect speech)."
 - She said, "I gave the key to my roommate (Indirect Speech)."

2 M BL1 CO4

4 Select the appropriate word and fill in the blanks. (1/2mx 2 = 1M)

- Let us _____ to the auditorium. (precede, proceed)
 - The bus _____ (fare/fair) increased drastically.
- Change the voice. (1/2m x 2 = 1M)
- The students have paid the fees.
 - She was congratulated by her friends.

2 M BL3 CO5

5 Rewrite the sentences correctly (1/2m x 4 = 2 M)

- I will look forward to see you soon.
- Please don't repeat the mistake again.
- The water is very hot to drink.
- Harsha don't know Hindi.

PART-B

5 X 10= 50M

5 M BL2 CO1

Note : Answer all questions

6 a. Summarize " On the Conduct of Life" by William Hazlitt briefly.

5 M BL3 CO1

- Use the following hints to write a paragraph, give it a title and a moral.
Hints : The King of Scotland was defeated six times-----hid himself in a cave saw a spider trying to climb up to the roof of the cave the spider fell six times and then succeeded- the king was encouraged - tried again to defeat his enemies and succeeded.

OR

Code :S222821

- c. Summarize the young boy's attitude towards his new school as explained in William Hazlitt's "On the Conduct of Life." 5 M BL2 CO1
- d. Examine the structure of writing an effective paragraph. 5 M BL3 CO1
- 7 a. Outline the central idea of the poem "If." 5 M BL2 CO2
- b. Describe compounding, clipping and blending words with examples. 5 M BL1 CO2
- OR
- c. Outline the poet's views on honesty and fortitude of character from the poem "If." 5 M BL1 CO2
- d. Describe briefly "Recent floods due to continuous rains." 5 M BL2 CO2
- 8 a. Discuss the dream that Martin Luther King envisions for America. 5 M BL2 CO3
- b. Describe your childhood memories. 5 M BL2 CO3
- OR
- c. Discuss "I have a Dream" as a plea for freedom and equality. 5 M BL2 CO3
- d. Describe your views on "Is country life better than city life." 5 M BL2 CO3
- 9 a. Examine the conflict of the Poem "The Road not Taken." 5 M BL3 CO4
- b. Justify the topic "Should the minimum voting age be raised/lowered/kept the same?" 5 M BL5 CO4
- OR
- c. Examine the spirit of regret and longing in the poem "The Road not Taken." 5 M BL3 CO4
- d. Justify the topic "Is online learning effective?" 5 M BL5 CO4
- 10 a. Explain the important features of a blog. 5 M BL2 CO5
- b. Create a blog on the topic "Writing captions on instagram." 5 M BL6 CO5
- OR
- c. Explain the process of writing "Tweets" and "Instagrams." 5 M BL2 CO5
- d. Create a tweet in response to the proposal of the "New Labour Code i.e. 4-day work per week, 12 working hours every working day in the week." 5 M BL6 CO5

Hall Ticket No-

Code S222825

Stanley College of Engineering and Technology for Women (A)

B.E (CSE/EEE/ECE/IT/CME/AI&DS) II Semester (Backlog) Examinations January-2023

Mathematics--II

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5X2=10M

- 1 Solve $(x^2 + y^2) dx + 2xy dy = 0$ 2 M L3
- 2 Find the C.F. of $(D^2 + D + 1)y = 0$ 2 M L2
- 3 Define Beta function and Gamma function 2 M L1
- 4 Find the Eigen Values of the Matrix $A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$ 2 M L3
- 5 Find $L\{t^2 \sin at\}$ 2 M L3

PART-B

Note: Answer all questions

5X10=50M

- 6 a. Solve $\frac{dy}{dx} + y \cos x = y^3 \sin 2x$ 5 M L3
- b. Find the Orthogonal trajectories of $\frac{x^2}{a^2 + \alpha} + \frac{y^2}{b^2 + \alpha} = 1$, α being the parameter. 5 M L3
- c. Solve $(1 + e^{xy}) dx + e^{xy} (1 - \frac{x}{y}) dy = 0$ 5 M L3
- d. Solve $p = \sin(y - xp)$. Also find its singular solutions. 5 M L3
- 7 a. Solve $(D^2 + 4)y = e^x + \sin 2x + \cos 2x$ 5 M L3
- b. Solve $y'' + y = \sec x$ by the Method of Variation of parameters. 5 M L3
- c. Solve $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + 2y = x \log x$ 5 M L3
- d. Solve $(D^2 - 4D + 3)y = 3e^x \cos 2x + 2xe^{3x}$ 5 M L3
- 8 a. State and prove the relation between Beta and Gamma Function. 6 M L4
- b. Evaluate $\int_0^2 x^3 \sqrt{2-x} dx$ using Beta And Gamma function. 4 M L5
- c. State and Prove Rodrigue's Formula. 6 M L4
- d. Evaluate $\int_0^a x^4 \sqrt{a^2 - x^2} dx$ 4 M L5

- 9 a. Verify Cayley – Hamilton Theorem for the Matrix 6 M L4

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \text{ and also find } A^{-1}$$

- b. Discuss the nature of the Q.F. $x^2 + 4xy + 6xz + 2yz - y^2 + 4z^2$ 4 M L4

OR

- c. Find Eigen Values and corresponding Eigen Vectors of 8 M L3

$$\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

- d. Using Cayley – Hamilton Theorem, find A^8 if $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ 2 M L3

- 10 a. Find the Laplace Transform of (i) $t e^{2t} \sin 3t$ (ii) $e^{-t} \cos^2 t$ 5 M L3

- b. Find the Inverse Laplace Transform of $\frac{2s+12}{s^2+6s+12}$ and $\frac{s}{(s+2)^2}$ 5 M L3

OR

- c. Find $L \{ t \cos at \}$ 2 M L5

- d. Using Laplace Transform method, solve 8 M L5
 $(D^2 + 1)y = 6 \cos 2t, t > 0$, If $y = 1, Dy = 1$ when $t = 0$.

Stanley College of Engineering and Technology for Women (A)

B.E (CSE/EEE/ECE/IT/CME/AI&DS) I Semester (Backlog) Examinations January-2023

Mathematics-I

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

1. Test the convergence of the series $\sum (1 + \frac{1}{n})^{-n^2}$ 2M
2. Verify Cauchy's mean value theorem for $f(x) = x^2$ and $g(x) = x^3$ in $[1, 2]$ 2M
3. If $u = \frac{x+y}{1-xy}$ and $v = \tan^{-1} x + \tan^{-1} y$ then prove that u and v are functionally dependent. 2M
4. Find grad f if $f = x^2y + y^2x + z^2$ at $(1, 1, -2)$ 2M
5. Find the first approximation of the root lying between 0 and 1 of the equation $x^3 + 3x - 1 = 0$ by Newton-Raphson formula. 2M

PART B

Note: Answer all questions

5 X 10 = 50M

6. (a) Discuss the convergence or divergence of $\frac{x}{1.2} + \frac{x^2}{2.3} + \frac{x^3}{3.4} + \dots$ 5M
- (b) Examine the convergence of the series $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$ 5M
- (OR)
- (c) Test the convergence of the series $1 + \frac{2}{5}x + \frac{6}{9}x^2 + \dots + \frac{2^n-2}{2^{n+1}}x^{n-1} + \dots$ ($x > 0$) 5M
- (d) Show that the exponential series $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ converges absolutely for all values of x . 5M
7. (a) Find the evolute of the parabola $y^2 = 4ax$ 5M
- (b) Verify Cauchy's Mean value theorem for the functions $f(x)$ and $f^1(x)$ in $(1, e)$ given $f(x) = \log x$ 5M

(OR)

- (c) Prove that $\frac{\pi}{6} + \frac{1}{5\sqrt{3}} < \sin^{-1}\left(\frac{3}{5}\right) < \frac{\pi}{6} + \frac{1}{8}$ 10M
8. (a) If $u = x + y + z$, $y + z = uv$, $z = uvw$ then show that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$ 5M
- (b) If $u = x^2 + y^2$, $x = at^2$, $y = 2at$ then find $\frac{du}{dt}$ 5M
- (OR)
- (c) A rectangular box open at the top is to have volume 32 cubic ft. Find the dimensions of the box requiring least material for its construction. 5M
- (d) Find the maximum and minimum values of $f(x,y) = x^3 + y^3 - 3axy$ 5M
9. (a) If $F = (5xy - 6x^2)\mathbf{i} + (2y - 4x)\mathbf{j}$ Evaluate $\int_c \vec{F} \cdot d\vec{r}$ where c is the curve in the xy plane $y = x^3$ from $(1,1)$ to $(2,8)$ 5M
- (b) Find $\text{div } \vec{f}$ when $\vec{f} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$ 5M
- (OR)
- (c) Verify Green's theorem in the plane for $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the region bounded by $y = \sqrt{x}$ and $y = x^2$ 10M
10. (a) Solve the following system of equations using Gauss Seidel iteration method
 $10x + 2y + z = 9$, $x + 10y - z = -22$, $-2x + 3y + 10z = 22$ 10M
- (OR)
- (c) Using Newton's divided difference formula evaluate $f(8)$ given that 10M

x	4	5	7	10	11	13
f(x)	48	100	294	900	1210	2028

Stanley College of Engineering and Technology for Women(A)**B.E(ECE & EEE) I Semester (Backlog) Examinations January-2023****Engineering Chemistry****Time: 3 hours****Max.Marks:60****PART-A****Note: Answer all questions (Compulsory)****5 X 2=10M**

- | | |
|---|------------|
| 1 What is cathodic protection? | 2 M L1 CO1 |
| 2 Define entropy and enthalpy. | 2 M L2 CO2 |
| 3 Calculate the bond order of NO molecule. | 2 M L3 CO3 |
| 4 Describe the significance of Octane number and Cetane number of a fuel. | 2 M L4 CO4 |
| 5 What are the composite materials? Write its classification. | 2 M L2 CO5 |

PART-B**Note: Answer all questions****5 X 10= 50M**

- | | |
|---|------------|
| 6 a. Explain the purification of water by ion exchange process and give its advantages,disadvantages. | 6 M L2 CO1 |
| b. Discuss the various (any three) factors affecting rate of corrosion. | 4 M L6 CO1 |

OR

- | | |
|--|------------|
| c. What are boiler troubles? Explain formation & prevention of scale and sludge. | 6 M L2 CO1 |
| d. Write a short note on galvanizing. | 4 M L1 CO1 |
| 7 a. Explain the physical significance of entropy. | 6 M L2 CO2 |
| b. Derive nernst equation and write its applications. | 4 M L6 CO2 |

OR

- | | |
|--|------------|
| c. Describe the construction and working of calomel electrode with a neat diagram. | 6 M L5 CO2 |
| d. The efficiency of an engine is 0.42. Calculate the heat that must be withdrawn from the reservoir at higher temperature to produce 203 cal of work. | 4 M L3 CO2 |
| 8 a. Draw the molecular orbital diagram of N ₂ and find out the bond order. | 6 M L4 CO3 |
| b. Describe the preparation and applications of Poly lactic acid. | 4 M L2 CO3 |

OR

- | | |
|---|------------|
| c. Discuss the preparation,properties and applications of PET and Kevlar | 6 M L6 CO3 |
| d. Analyze the doping mechanism of conduction in poly acetylene(p-doping & n-doping) | 4 M L4 CO3 |
| 9 a. Explain proximate analysis of coal? How is it carried out ? What is its significance? | 6 M L1 CO4 |
| b. A sample of coal has the following composition: C=83% ,H=7.5% , S=1.5% , N=0.6 % ,O=8.4 % find the gross calorific value using Dulong Formula. | 4 M L3 CO4 |

OR

- | | |
|---|------------|
| c. Explain fractional distillation of petroleum with a neat diagram. | 6 M L2 CO4 |
| d. Explain the concept of trans esterification of biodiesel. | 4 M L2 CO4 |
| 10 a. What are Lithium ion batteries? Explain the construction,working and applications of Lithium ion batteries. | 6 M L5 CO5 |
| b. Explain the construction and working of CH ₃ OH-O ₂ fuel cell and write its advantages. | 4 M L2 CO5 |

OR

- | | |
|--|------------|
| c. Illustrate the principles of green chemistry. | 6 M L3 CO5 |
| d. Make the use of Nanotechnology write a short note on the following a) Carbon nano tubes b) Nano crystals. | 4 M L3 CO5 |

Stanley College of Engineering and Technology for Women(A)**B.E(EEE & ECE) I Semester Backlog Examinations-January-2023****Fundamentals of Electrical Engineering**

Time: 3 hours

Max.Marks:60

Part-A**Note: Answer all questions (Compulsory)****5 X 2=10M**

- 1 State Kirchhoff's current law. 2 M
- 2 Define Reactive Power and Apparent Power. 2 M
- 3 What are the differences between statically induced emf & dynamically induced emf? 2 M
- 4 Why single-phase induction motor is not self-starting? 2 M
- 5 Mention any two components of LT Switchgear. 2 M

Part-B**Note: Answer all questions****5 X 10= 50M**

- 6
 - a. Formulate the expressions for equivalent resistance if 'n' number of resistances is connected in series and parallel combinations. 5 M
 - b. Explain Thevenin Theorem with one example. 5 M
- OR**
- c. Explain Norton Theorem with one example. 5 M
 - d. What is Superposition Theorem, Justify with an example? 5 M
- 7
 - a. Derive the expressions for resistances when a network is converted from star to delta. 5 M
 - b. The equation of an alternating current is $I=42.42 \sin 628t$. Determine (i) its maximum value (ii) Frequency (iii) RMS value (iv) Average value and (v) Form Factor. 5 M
- OR**
- c. Explain the behavior of a pure inductor when connected across a single-phase ac supply. 5 M
 - d. Show that the line voltage is $\sqrt{3}$ times the phase voltage in a star connected 3- ϕ system. 5 M
- 8
 - a. Explain the principle of operation of a three-phase Induction Motor. 5 M
 - b. A 10 KVA, 500/250V single phase transformer has the following results
 - i) O.C. Test : 250V, 0.8A, 200W
 - ii) S.C. Test : 120V, 20A, 500W
 Calculate the Efficiency at Full load with 0.8P.F. lagging 5 M

- c. Explain the principle of operation of a Transformer. 5 M
- d. Discuss in detail about the Rotating Magnetic Field in a 3-phase Induction Motor. 5 M
- 9 a. With a neat circuit diagram, Discuss about the Split-phase type Induction Motor 5 M
- b. Derive an expression for the EMF Equation of a DC Generator. 5 M
- OR**
- c. A 6-pole machine has an armature with 90 slots and 8 conductors per slot, the flux per pole is 0.05 wb and speed at 1000 rpm. Determine induced emf if winding is (i) lap connected and (ii) wave connected. 5 M
- d. Draw a neat sketch of a DC Machine and explain the function of various parts. 5 M
- 10 a. Explain the concept of earthing. 5 M
- b. Clearly illustrate types of wires and cables. 5 M
- OR**
- c. Explain in detail MCB and ELCB. 5 M
- d. Mention methods of power Factor improvement. 5 M

Stanley College of Engineering and Technology for Women(A)

B.E (CSE/EEE/ECE/IT/CME/AI&DS) I Semester (Backlog) Examinations January-2023

Programming for Problem Solving

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 Define implicit type casting with example. 2 M
- 2 Differentiate between 'break' and 'continue' statements. 2 M
- 3 Compare call by value and call by reference. 2 M
- 4 What are the bit fields in C? 2 M
- 5 What is the use of fseek() function in files. Write its syntax? 2 M

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. Discuss about different computer languages with examples. 5 M
- b. What is an operator? List and explain various types of operators. 5 M

OR

- c. Explain about the basic data types in c language with example. 5 M
- d. Draw the flowchart and write a C program to compute simple interest. 5 M
- 7 a. Explain the for loop and nested for loop with suitable examples. 5 M
- b. Write a C program to find the sum of first and last digit of a number. 5 M

OR

- c. Explain switch statement with syntax and example. 5 M
- d. Implement a C program to find the reverse of an integer number and check whether it is palindrome or not. 5 M
- 8 a. What is an array? What are the different ways of initializing the arrays? 5 M
- b. Explain different string handling functions with syntaxes. 5 M

OR

- c. Explain how arguments are passed to a function using references? 5 M
- d. Demonstrate storage classes available in C language with examples. 5 M
- 9 a. What is Structure? Explain initialization and accessing structures. 5 M
- b. What are enumeration? Give an example to explain the same. 5 M

OR

- c. Explain the concept of Nested structures with a sample C program. 5 M
- d. What is dynamic memory management? Explain different dynamic memory management functions available in C language. 5 M
- 10 a. Explain command line arguments with an example. 5 M
- b. Write a program to perform Linear Search operation. 5 M

OR

- c. Demonstrate file types and file opening modes. 5 M
- d. Explain bubble sort operation with an example 5 M

Hall Ticket No-

Code: S22325

Stanley College of Engineering and Technology for Women(A)

B.E (ECE,EEE & IT) I Semester Backlog Examinations-January-2023

Environmental Science

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- | | | | |
|---|--|----|----|
| 1 | List the effects of deforestation on tribal people. | 2M | L1 |
| 2 | Discuss the two channel energy flow in an ecosystem. | 2M | L5 |
| 3 | List the various bio geographical zones in India. | 2M | L1 |
| 4 | List the salient features of air act. | 2M | L1 |
| 5 | Define global warming. | 2M | L1 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | | | |
|----|--|----|----|
| 6 | a. Discuss the various ill-effects of modern agriculture. | 5M | L5 |
| | b. Relate soil erosion impact desertification and fertility. | 5M | L1 |
| | OR | | |
| | c. Discuss soil erosion and desertification. | 5M | L5 |
| | d. Explain problems of Dam. | 5M | L3 |
| 7 | a. Discuss the function of an ecosystem. | 5M | L5 |
| | b. Explain about desert ecosystem. | 5M | L3 |
| | OR | | |
| | c. Analyze the structure of Ecosystem. | 5M | L4 |
| | d. Explain about Types of Ecological pyramids. | 5M | L4 |
| 8 | a. Explain about endangered and endemic species of India. | 5M | L3 |
| | b. Describe the conservation of biodiversity. | 5M | L5 |
| | OR | | |
| | c. Write the major threats to biodiversity. | 5M | L1 |
| | d. Relate the Values of Biodiversity. | 5M | L1 |
| 9 | a. Explain the causes of water pollution. | 5M | L1 |
| | b. Write about Wildlife Act and Forest conservation Act. | 5M | L1 |
| | OR | | |
| | c. Explain about Environmental Legislation. | 5M | L1 |
| | d. Discuss effects and control measures of Air pollution. | 5M | L5 |
| 10 | a. Analyze mitigation measures for Ozone layer depletion. | 5M | L4 |
| | b. Explain about disaster management of India. | 5M | L1 |
| | OR | | |
| | c. Explain about Acid rains. | 5M | L2 |
| | d. Categorize Types of Disasters. | 5M | L3 |